

Grade 12
Pre-Calculus Mathematics
Achievement Test

Booklet I

January 2025

Grade 12 Pre-Calculus Mathematics Achievement Test:
Booklet 1 (January 2025)

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While the department is committed to making its publications as accessible as possible, some parts of this document are not fully accessible at this time.

Available in alternate formats upon request.

Grade 12 Pre-Calculus Mathematics Achievement Test

DESCRIPTION

Time Required to Complete the Test: 3 hours

Additional Time Allowed: 30 minutes

Number and Marks by Question Type

	Selected Response	Constructed Response	Marks
Booklet 1*	—	16	40
Booklet 2	9	19	50
Total	9	35	90

* The first 5 questions with the symbol  in Booklet 1 require a scientific calculator. You will have access to your calculator for the first 45 minutes of the test.

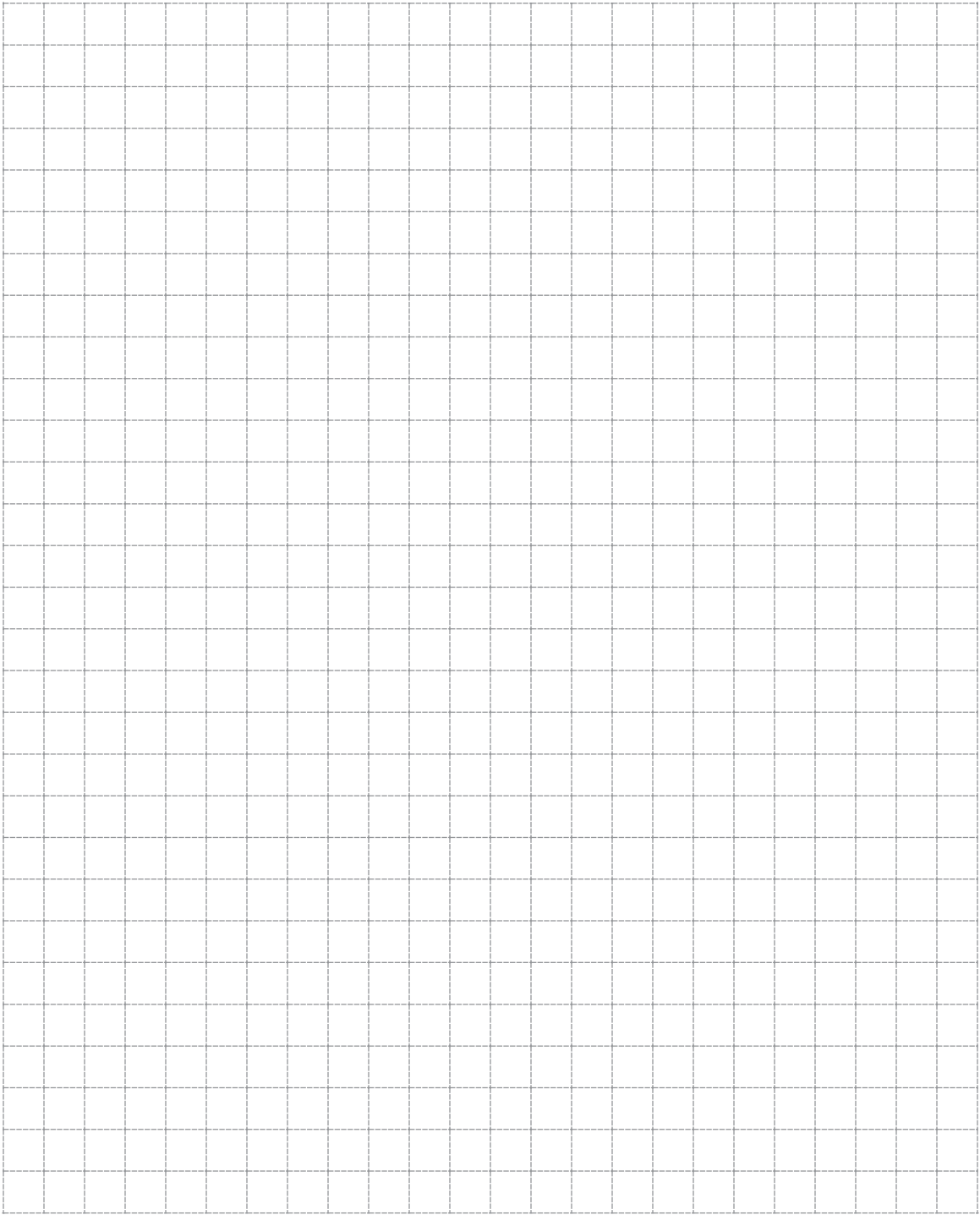
Note that diagrams and graphs provided in the test booklets may not be drawn to scale.

DIRECTIONS

- Write each solution in the space provided.
- For full marks, your answers must show all pertinent diagrams, calculations, and explanations.
- Your solutions should be neat, organized, and clear.
- Some answers are to be given as decimal values. Rounding too early in your solution may result in an inaccurate final answer for which full marks will not be given.
- Express your answers as exact values or correct to the nearest thousandth (3 decimal places) unless instructed otherwise.

Electronic communication between students through phones, email, or file sharing during the test is strictly prohibited. Please turn off your cell phone and all other such devices.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



Formula Sheet

$$s = \theta r$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha$$

$$\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$$

$$\cos 2\alpha = 1 - 2 \sin^2 \alpha$$

$$\cos 2\alpha = 2 \cos^2 \alpha - 1$$

$$\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha}$$

$$\log_a(MN) = \log_a M + \log_a N$$

$$\log_a\left(\frac{M}{N}\right) = \log_a M - \log_a N$$

$$\log_a(M^n) = n \log_a M$$

$$P(n, r) \text{ or } {}_nP_r = \frac{n!}{(n-r)!}$$

$$C(n, r) \text{ or } {}_nC_r = \frac{n!}{r!(n-r)!}$$

$$t_{k+1} = {}_nC_k a^{n-k} b^k$$

$$\text{Given } ax^2 + bx + c = 0,$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Terminology Sheet

Some questions may contain directing words such as *explain, identify, and justify*. These words are defined below.

Describe: Use words to provide the process or to report details of the response.

Determine: Use a mathematical formula, an algebraic equation, or a numerical calculation to solve a problem.

Evaluate: Find the numerical value.

Explain: Use words to provide the cause of or reason for the response, or to render the response more clear and understandable.

Identify: Recognize and select the answer by stating or circling it.

Justify: Show reasons for or give facts that support a position by using mathematical computations, words, and/or diagrams.

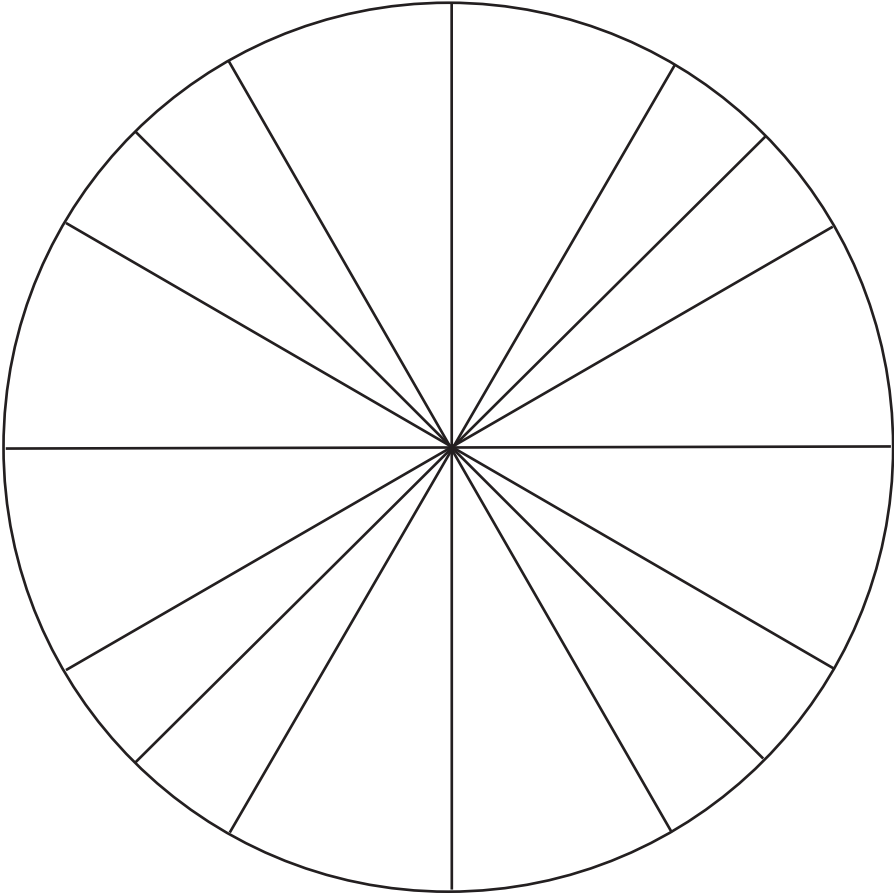
Sketch the graph: Provide a detailed drawing with key features of the graph that includes a minimum of 2 coordinate points.

Solve: Give a solution for a problem or determine the value(s) of a variable.

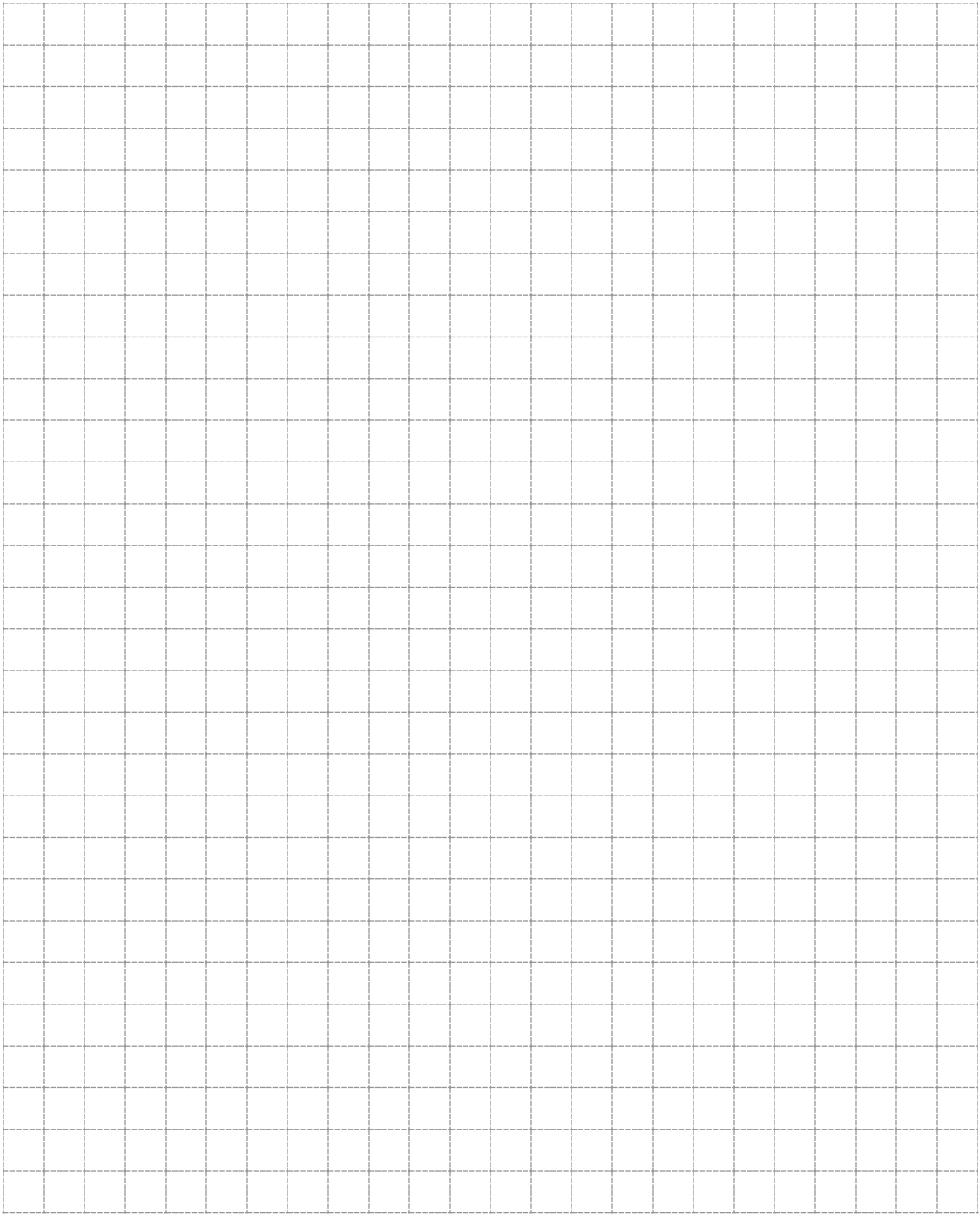
State: Give an answer without an explanation or justification.

Verify: Establish the truth of a statement by substitution or comparison.

NO MARKS WILL BE AWARDED FOR WORK DONE ON THIS PAGE.



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Question 1 **1 mark**

101

Youyi wants to buy a new cell phone from a company that offers 33 different types of cell phones. Each cell phone is available in 5 different colours. Determine the total number of available options.

Question 2 **2 marks**

102

Determine the central angle subtended by an arc length of 30 cm on a circle with a diameter of 20.6 cm. Express your answer in degrees.

Question 3 **3 marks**

103

A choir has 14 sopranos. Karin and Ellen are two of them. Determine the number of possible ensembles of 3 sopranos that could be formed if either Karin or Ellen, or both, must be included.

Question 4 **3 marks**

104

Loïc invests \$5000 at an annual interest rate of 6.75% compounded semi-annually. The value of his investment grows according to the following formula:

$$A = P \left(1 + \frac{r}{n} \right)^{nt}$$

where A is the value of the investment after t years

P is the amount of the initial deposit

r is the annual interest rate (as a decimal)

n is the number of compounding periods per year

t is the time in years

Determine, algebraically, the length of time it will take Loïc's investment to grow to \$9500.

Question 5 **4 marks**

105

Solve, algebraically, over the interval $[0, 2\pi]$.

$$\csc^2 \theta - \csc \theta - 6 = 0$$

Note: A calculator is not required for the remaining test questions.

Question 6

2 marks

106

Determine how many 4-digit numbers can be made using the digits 0, 1, 2, 3, 4, and 5, if the third digit must be 3, and repetition is not allowed.

Question 7

3 marks

107

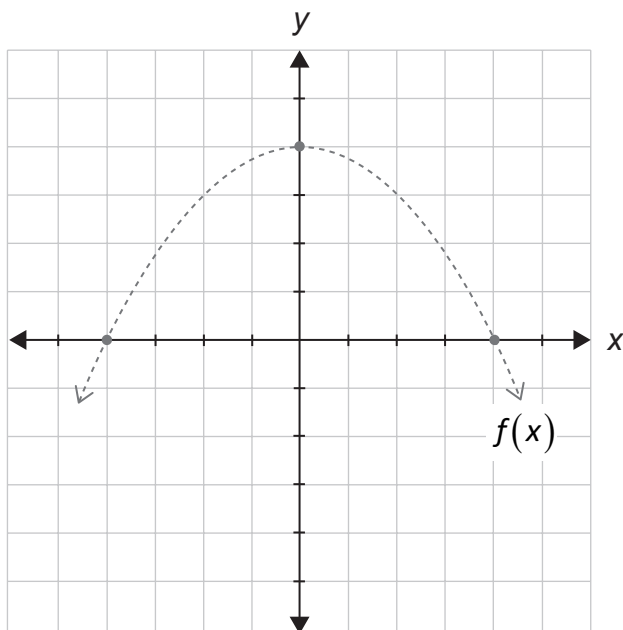
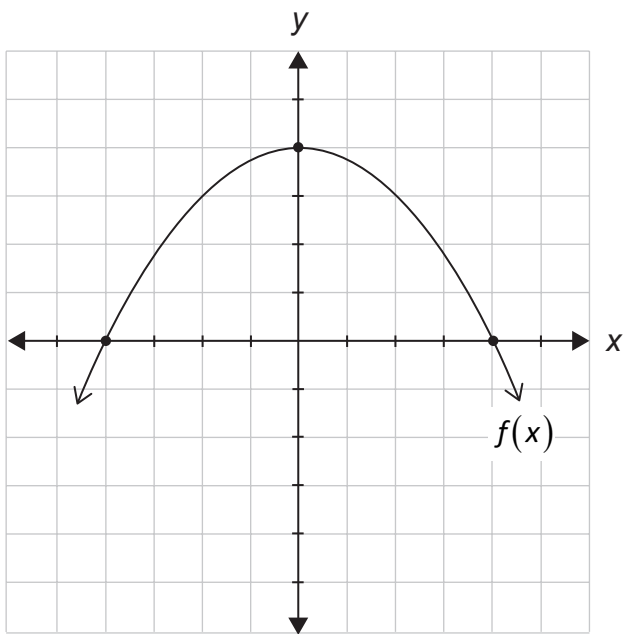
Describe the transformations required to obtain the graph of the function $y = -4f\left(\frac{1}{3}x\right)$ from the graph of $y = f(x)$.

Question 8

2 marks

108

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 9

1 mark

109

Describe how to determine if $(x - a)$ is a factor of the polynomial function, $p(x)$.

Question 10

3 marks

110

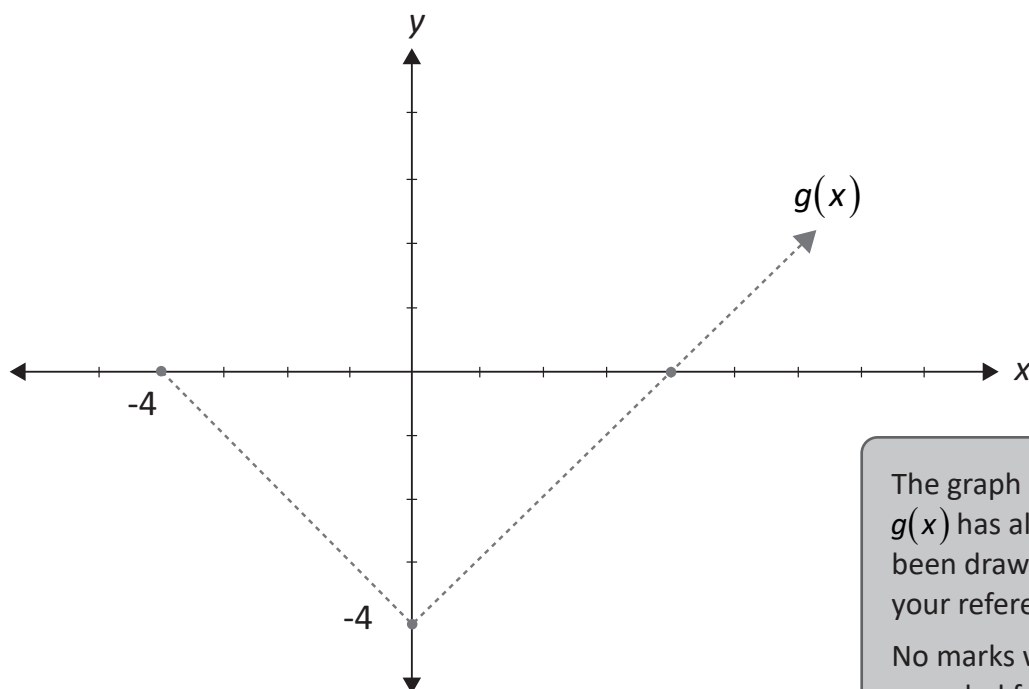
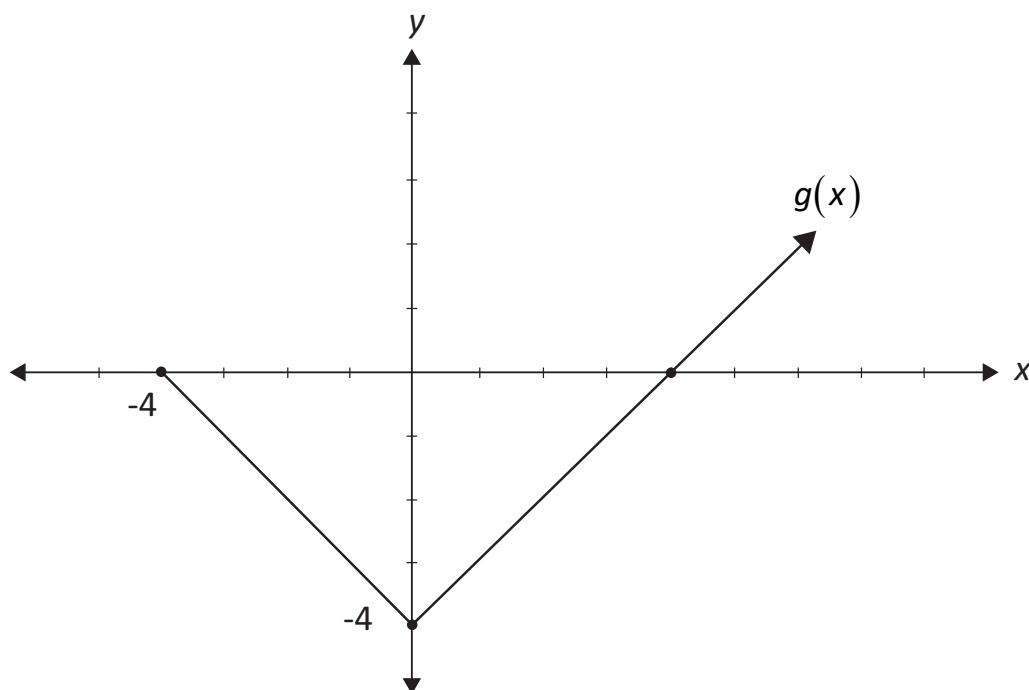
Express $\frac{1}{2}\log x - 2\log y + \log z$ as a single logarithm.

Question 11

3 marks

111

Given the graph of $g(x)$, sketch the graph of $f(x) = \frac{1}{g(x)}$.



The graph of $g(x)$ has already been drawn for your reference.
No marks will be awarded for the graph of $g(x)$.

Question 12

a) 1 mark b) 3 marks

112
113

Given the identity $\frac{1 - \tan^2 \theta}{\sec^2 \theta} = \cos 2\theta$,

a) state a non-permissible value of θ .

b) prove the identity for all permissible values of θ .

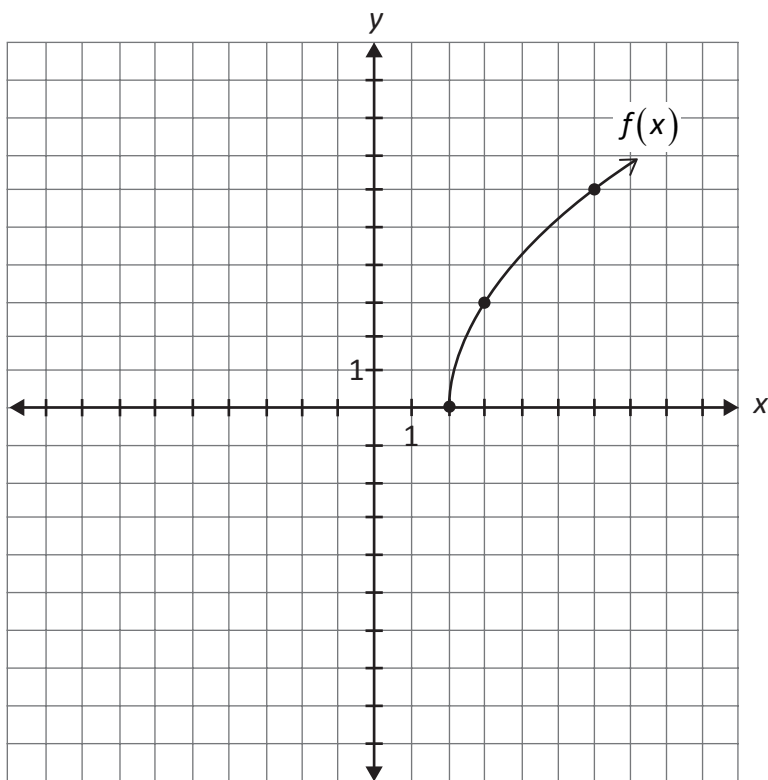
Left-Hand Side	Right-Hand Side

Question 13

2 marks

114

Determine an equation of the radical function represented by the graph of $f(x)$.



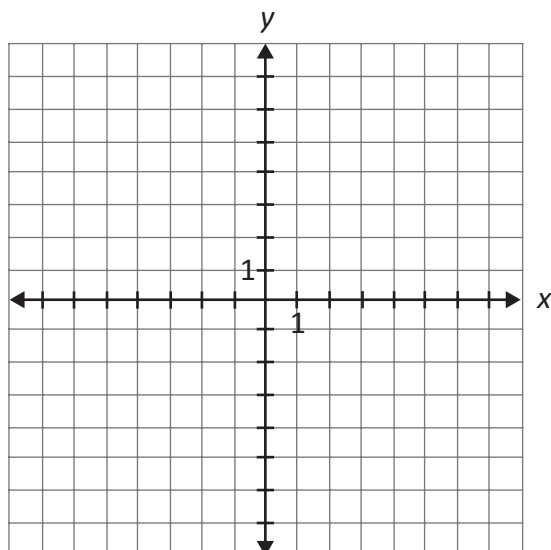
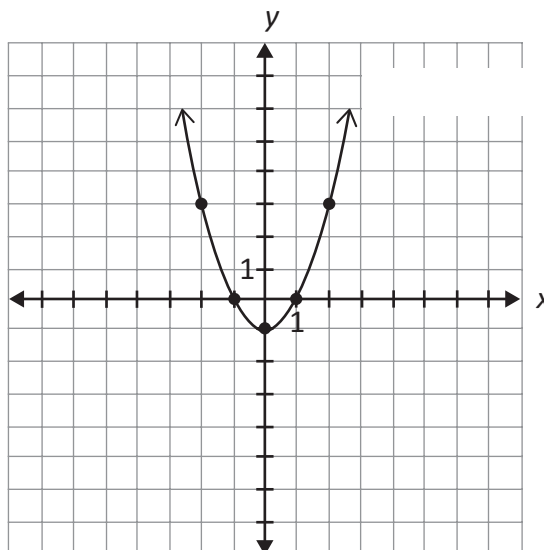
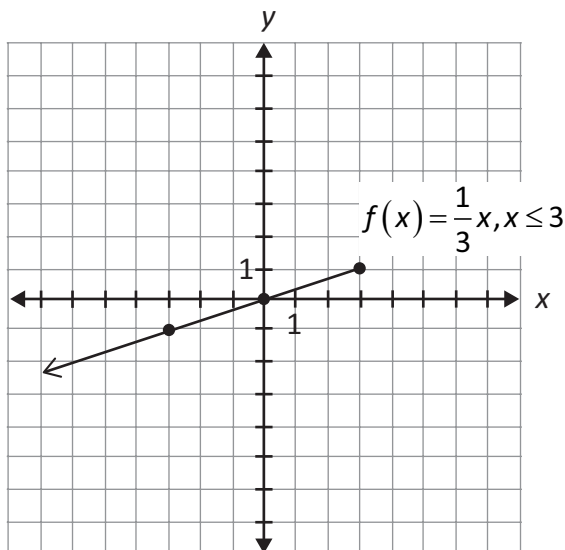
$f(x) =$ _____

Question 14

2 marks

115

Given the graphs of $f(x)$ and $g(x)$, sketch the graph of $g(f(x))$.



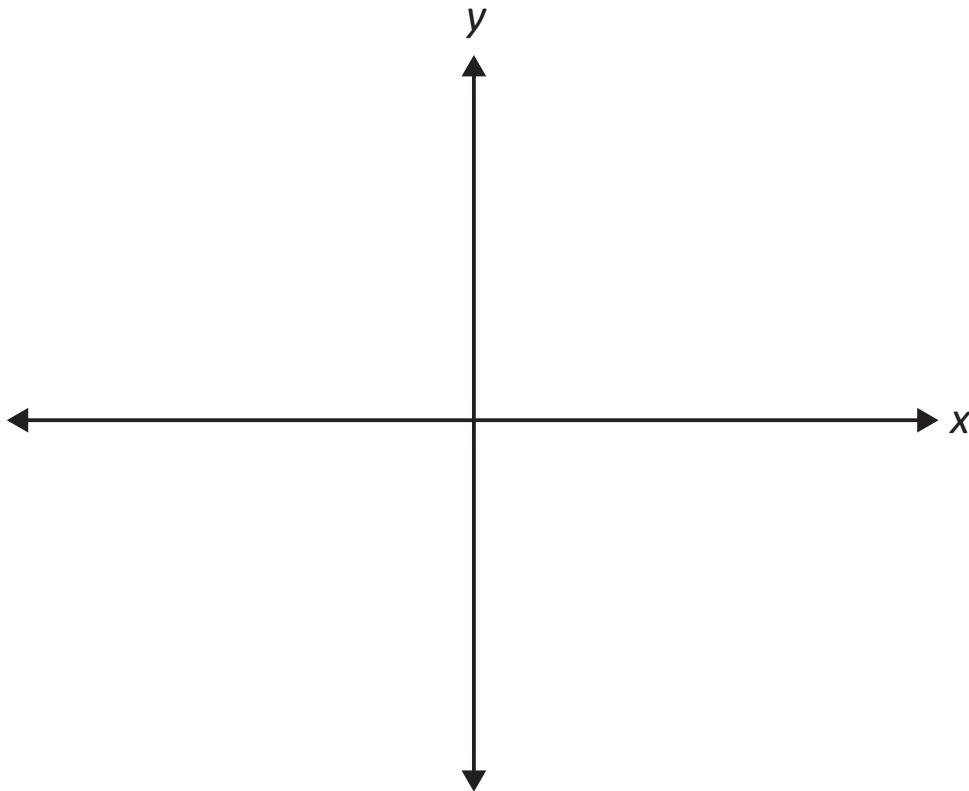
Question 15

3 marks

116

Sketch a graph of $p(x)$ that satisfies all of the following conditions:

- $p(x)$ is a polynomial function of degree 3
- $p(x)$ has a zero at -5 with a multiplicity of 2
- $p(x)$ has a zero at 1
- $p(x)$ has a leading coefficient of -2



Question 16

2 marks

117

Sketch the graph of $f(x) = \frac{x^2 + x}{x}$.

